

Structural failure
Excessive deflection
Over designed structure

Cantilever

$$V(x) = -(wL + P) + w \cdot x$$

$$M(x) = -M_{\text{base}} + R_y \cdot x - w \frac{x^2}{2}$$

EMT 3203: Solid Mechanics II

B.Sc. Mechatronics Engineering

Continuous Assessment Test II

2026

Instructions:

- ✧ Answer all questions. The test has a total of 30 marks.
- ✧ You have 1 hour to complete this test.
- ✧ Write your name, registration number, and your answers in the provided answer booklet.

Question 1 (10 marks)

A printed circuit board (PCB) is mounted between two fixed supports in an electronics rack. The PCB has dimensions $L = 200$ mm, $b = 100$ mm, $h = 1.6$ mm, carries a total component weight of 12 N (well distributed), and is made of FR4 composite ($E = 20$ GPa). The uniformly distributed load intensity is $w = 60$ N/m.

- (a) Calculate the second moment of area of the PCB. [2 marks]
- (b) Determine the maximum midspan deflection. [3 marks]
- (c) Check whether the deflection satisfies the standard criterion ($\delta_{\text{max}} < L/100$). [2 marks]
- (d) Propose two design modifications to stiffen the PCB and analyze their effect on deflection. [3 marks]

Question 2 (10 marks)

A lightweight quadcopter drone has hollow carbon fiber arms supporting motors and propellers. Each arm acts as a cantilever (fixed at the drone body, free at the motor end) and must resist both rotor thrust and motor torque.

The arm geometry and material are:

- Hollow circular tube: Outer diameter $D = 20$ mm, inner diameter $d = 16$ mm, length $L = 300$ mm.
- Section properties: $I = 4618$ mm⁴, $J = 9236$ mm⁴, $c = 10$ mm.
- Material: Carbon fiber composite, yield strength $\sigma_{\text{yield}} = 400$ MPa, target safety factor 3–5.

$$\frac{bh^3}{12}$$

$$\frac{bh^3}{12} = \frac{20 \times 10^3}{12} = 1666.67 \text{ mm}^4$$

P $\frac{PL^3}{192EI}$ $\frac{PL^3}{3EI}$ $\frac{PL^3}{16EI}$
 UDL $\frac{WL^4}{384EI}$ $\frac{5WL^4}{384EI}$

Loading:

- Rotor thrust $F = 50$ N downward at free end.
- Motor torque $T = 3$ N-m along arm axis.

$$V(x) = W_2 \cdot (xL + P)$$

$$M_{OD} = R_y \cdot x - (M_{max} + \frac{W_2 x^2}{2})$$

- Compute the bending moment and bending stress at the fixed end. [3 marks]
- Compute the torsional shear stress due to motor torque at the fixed end. [2 marks]
- Determine the von Mises equivalent stress combining bending and torsion effects. [3 marks]
- Evaluate whether the arm meets the safety factor requirement under combined loading. [2 marks]

Question 3 (10 marks)

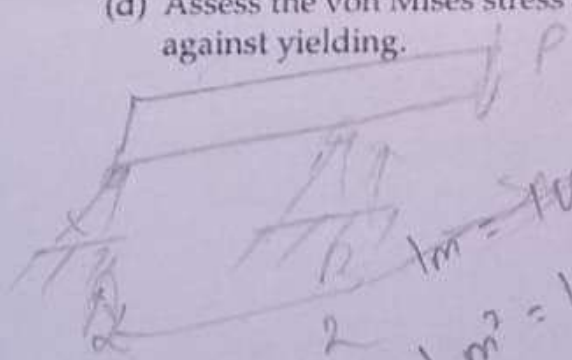
The main shaft of a horizontal-axis wind turbine transmits power from the rotor to the gearbox while supporting the rotor weight and shaft self-weight. The shaft is a hollow steel tube ($D = 100$ mm, $d = 80$ mm, $L = 2.0$ m), with a pinned support at bearing A and a roller support at bearing B. The rotor assembly overhangs beyond bearing B.

Loading

- Point load $P_1 = 1200$ N at the rotor tip.
- Uniformly distributed load $W = 300$ N/m along the shaft.
- Constant transmitted torque $T = 5000$ N-m along the shaft.

Cross-section properties: $I = 2.896 \times 10^6$ mm⁴, $J = 5.792 \times 10^6$ mm⁴, $c = 50$ mm. Material: AISI 1045 steel, $\sigma_{yield} = 310$ MPa, safety factor target 2.5–4.0.

- Calculate the support reactions at bearings A and B. [2 marks]
- Determine the location and magnitude of maximum bending moment. [3 marks]
- Compute bending and torsional stresses at the critical section. [3 marks]
- Assess the von Mises stress and evaluate the safety factor against yielding. [2 marks]



$1 \text{ m} = 1000 \text{ mm}$
 $1 \text{ m}^2 = 1 \times 10^6 \text{ mm}^2$
 $M_B = (B P_L) \cdot P_1 + (B P_{L/2} \times W L$

End of Questions